

# **Data Structures**



**Submitted to**

**Mr. Bilal Bukhari**

**Submitted by**

**FAREEHA BATOOL**

**FA25-BCS-017**

**COMSATS UNIVERSITY ISLAMABAD  
(ATTOCK CAMPUS)**

## LAB TASK 01:

### CODE:

```
#include <iostream>

using namespace std;

int main()
{
    int x, start, stop;

    int sum = 0;

    cout << "Enter value of X: ";
    cin >> x;

    cout << "Enter starting value: ";
    cin >> start;

    cout << "Enter stopping value: ";
    cin >> stop;

    for (int i = start; i <= stop; i++)
    {
        sum = sum + (x * x);
    }

    cout << "Sum = " << sum;
```

```
    return 0;
}
```

### Output:

```
Enter X:4
Enter starting value:2
Enter stopping value:5
Sum=64
-----
Process exited after 11.68 seconds with return value 0
Press any key to continue . . . |
```

### Lab task 02:

#### Code:

```
#include <iostream>

using namespace std;
```

```
class ArrayList
```

```
{
```

```
private:
```

```
    int arr[100];
```

```
    int size;
```

```
public:
```

```
    ArrayList()
```

```
{
```

```
    size = 0;  
}
```

// 1. Insert at end

```
void insertEnd(int val)
```

```
{  
    if (size < 100)  
    {  
        arr[size] = val;  
        size++;  
  
        cout << "Inserted " << val << " at end\n";  
    }  
    else  
    {  
        cout << "List full!\n";  
    }  
}
```

// 2. Insert at start

```
void insertStart(int val)
```

```
{  
    if (size < 100)  
    {  
        for (int i = size; i > 0; i--)  
        {
```

```
        arr[i] = arr[i - 1];
    }

    arr[0] = val;
    size++;

    cout << "Inserted " << val << " at start\n";
}
else
{
    cout << "List full!\n";
}
}
```

// 3. Insert after specific value

void insertAfter(int after, int val)

```
{
    int pos = -1;

    for (int i = 0; i < size; i++)
    {
        if (arr[i] == after)
        {
            pos = i;
            break;
        }
    }
```

```
}

if (pos == -1)
{
    cout << after << " not found!\n";
    return;
}

if (size < 100)
{
    for (int i = size; i > pos + 1; i--)
    {
        arr[i] = arr[i - 1];
    }

    arr[pos + 1] = val;
    size++;

    cout << "Inserted " << val << " after " << after << endl;
}
else
{
    cout << "List full!\n";
}
}
```

// 4. Insert before specific value

void insertBefore(int before, int val)

{

int pos = -1;

for (int i = 0; i < size; i++)

{

if (arr[i] == before)

{

pos = i;

break;

}

}

if (pos == -1)

{

cout << before << " not found!\n";

return;

}

if (size < 100)

{

for (int i = size; i > pos; i--)

{

arr[i] = arr[i - 1];

}

```
    arr[pos] = val;
    size++;

    cout << "Inserted " << val << " before "
        << before << endl;
}
else
{
    cout << "List full!\n";
}
}
```

// 5. Display

void display()

```
{
    if (size == 0)
    {
        cout << "List is empty!\n";
        return;
    }
}
```

cout << "List: ";

for (int i = 0; i < size; i++)

```
{
```

```
        cout << arr[i] << " ";  
    }
```

```
    cout << endl;  
}
```

// 6. Delete from end

void deleteEnd()

```
{  
    if (size == 0)  
    {  
        cout << "List empty!\n";  
        return;  
    }  
}
```

```
    cout << "Deleted " << arr[size - 1]  
        << " from end\n";
```

```
    size--;  
}
```

// 7. Delete from start

void deleteStart()

```
{  
    if (size == 0)  
    {
```

```
    cout << "List empty!\n";  
    return;  
}
```

```
cout << "Deleted " << arr[0]  
    << " from start\n";
```

```
for (int i = 0; i < size - 1; i++)  
{  
    arr[i] = arr[i + 1];  
}
```

```
size--;  
}
```

// 8. Delete specific value

```
void deleteVal(int val)  
{  
    int pos = -1;  
  
    for (int i = 0; i < size; i++)  
    {  
        if (arr[i] == val)  
        {  
            pos = i;  
            break;  
        }  
    }  
}
```

```
}  
}
```

```
if (pos == -1)  
{  
    cout << val << " not found!\n";  
    return;  
}
```

```
for (int i = pos; i < size - 1; i++)  
{  
    arr[i] = arr[i + 1];  
}
```

```
size--;
```

```
    cout << "Deleted " << val << endl;  
}  
};
```

```
int main()  
{  
    ArrayList list;
```

```
    int choice, val, target;
```

```
do
{
    cout << "\n===== MENU =====" << endl;
    cout << "1. Insert at end" << endl;
    cout << "2. Insert at start" << endl;
    cout << "3. Insert after specific value" << endl;
    cout << "4. Insert before specific value" << endl;
    cout << "5. Display list" << endl;
    cout << "6. Delete from end" << endl;
    cout << "7. Delete from start" << endl;
    cout << "8. Delete specific value" << endl;
    cout << "9. Exit" << endl;

    cout << "Choice: ";
    cin >> choice;

    switch (choice)
    {
    case 1:
        cout << "Enter value: ";
        cin >> val;

        list.insertEnd(val);
        break;

    case 2:
```

```
cout << "Enter value: ";
```

```
cin >> val;
```

```
list.insertStart(val);
```

```
break;
```

case 3:

```
cout << "Enter after value: ";
```

```
cin >> target;
```

```
cout << "Enter new value: ";
```

```
cin >> val;
```

```
list.insertAfter(target, val);
```

```
break;
```

case 4:

```
cout << "Enter before value: ";
```

```
cin >> target;
```

```
cout << "Enter new value: ";
```

```
cin >> val;
```

```
list.insertBefore(target, val);
```

```
break;
```

case 5:

list.display();

break;

case 6:

list.deleteEnd();

break;

case 7:

list.deleteStart();

break;

case 8:

cout << "Enter value to delete: ";

cin >> val;

list.deleteVal(val);

break;

case 9:

cout << "Exiting...\n";

break;

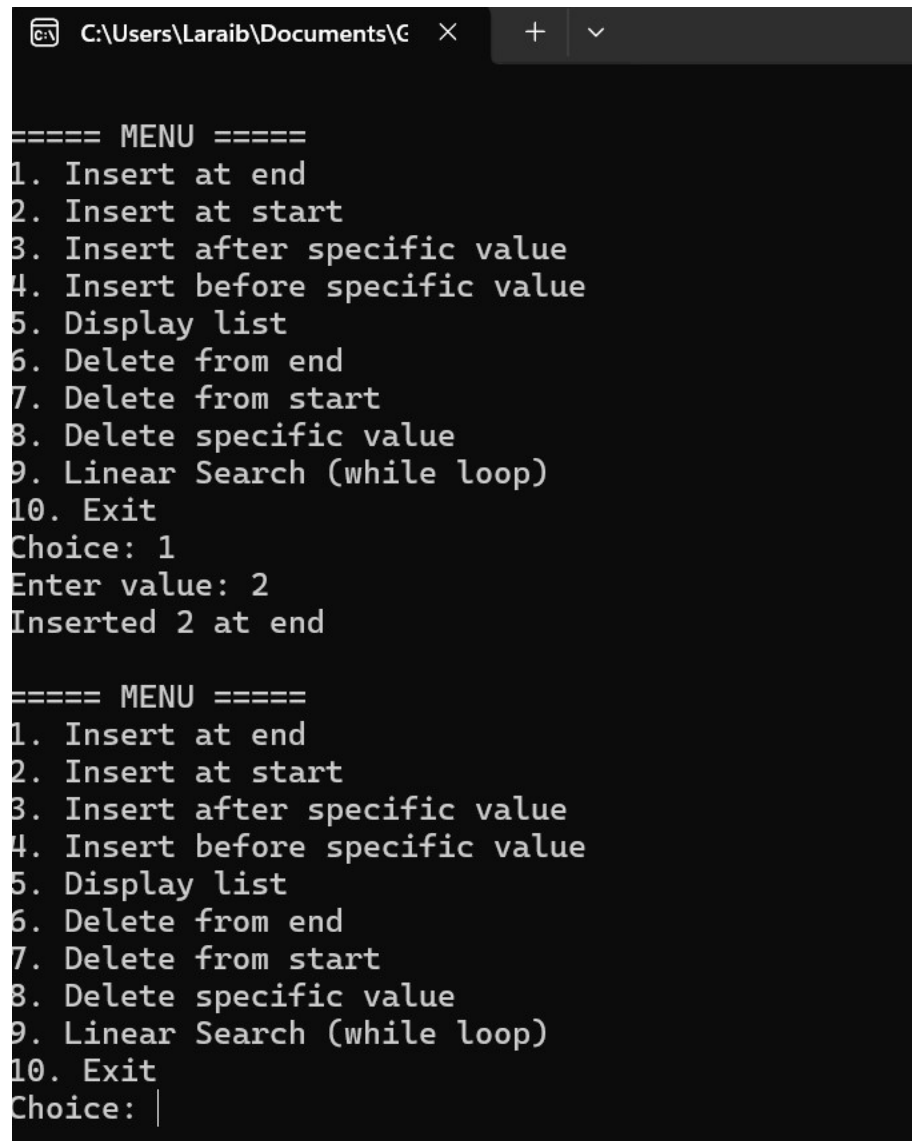
default:

cout << "Invalid choice!\n";

}

```
} while (choice != 9);  
  
return 0;  
}
```

### Output:



```
C:\Users\Laraib\Documents\G >  
===== MENU =====  
1. Insert at end  
2. Insert at start  
3. Insert after specific value  
4. Insert before specific value  
5. Display list  
6. Delete from end  
7. Delete from start  
8. Delete specific value  
9. Linear Search (while loop)  
10. Exit  
Choice: 1  
Enter value: 2  
Inserted 2 at end  
  
===== MENU =====  
1. Insert at end  
2. Insert at start  
3. Insert after specific value  
4. Insert before specific value  
5. Display list  
6. Delete from end  
7. Delete from start  
8. Delete specific value  
9. Linear Search (while loop)  
10. Exit  
Choice: |
```

### Lab task 03:

#### Code:

```
#include <iostream>

using namespace std;

int main()
{
    int arr[100], size, search, i;

    cout << "Enter number of elements: ";
    cin >> size;

    cout << "Enter " << size << " elements: ";

    for (i = 0; i < size; i++)
        cin >> arr[i];

    cout << "Enter value to search: ";
    cin >> search;

    i = 0;

    while (i < size && arr[i] != search)
        i++;

    if (i < size)
        cout << search << " found at index " << i << endl;
    else
```

```
cout << search << " not found!" << endl;
```

```
return 0;
```

```
}
```

### Output:

```
Enter number of elements: 4
Enter 4 elements: 2 3 5 7
Enter value to search: 5
5 found at index 2

-----
Process exited after 20.25 seconds with return value 0
Press any key to continue . . . |
```